WELDING OF UNS 31254 ALLOY

FWEL CONTRACT No : **1-14-16601/19**

CONTRACT NO. : **POGC-760-81-156**

PROJECT : **SOUTH PARS GAS FIELD DEVELOPMENT (PHASES 9&10)**

COMPANY : **NATIONAL IRANIAN OIL COMPANY**

SITE : **ASSALUYEH / SOUTH PARS GAS FIELD PERSIAN GULF, IRAN**

3	12/11/03	Issued for FEED Package	FWEL	D. PASS	D. PASS		Dayd Pass
2	17/10/03	Final Issue	FWEL	D. PASS	D. PASS		D. PASS
1	07/08/03	Issued for Company Review	FWEL	D. PASS	D. PASS		D. PASS
0	31/01/01	Issued for Basic Design Package	TECHNIP	F. DUROT	G. BROYE		F. DUROT
REV.	**DATE**	**DESCRIPTION**	**ORIG.**	**PRPD**	**CHKD**	**APP'D**	**CONTRACTOR's APP'D**

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GENERAL STATEMENT

The attached TOTAL EXPLORATION PRODUCTION (TEP) General Specification n° SP - TCS - 615 Rev. 2 is confirmed as the Project Specification, except as amended herein, and renumbered/reissued as specification n° RP 6340S 999 6300 005 Rev. 3 for SOUTH PARS PHASES 9 & 10 Project. △3

ADDENDUM

The following items refer to the attached TEP General Specification n° SP - TCS - 615 Rev. 2.
The clauses set out below shall modify or replace the existing clauses in the original specification as noted.

GENERAL COMMENTS

- The following terms which are used in this general Job Specification are deleted and replaced by :
- Engineering company by Contractor : shall mean the party which has entered under a Contract Agreement with POGC for the design, engineering, procurement and construction of the Onshore Facilities regarding the South Pars Field Development Project, Phases 9 & 10.
- TOTAL : shall be read as POGC or his nominated Representative.
- Manufacturer : shall mean any person, firm or company which manufacture or supply equipment or services for the performance of any item of the work.
- Work : shall mean works and / or services entrusted by contractor under the Contract with company..
- Correct the following definition :
- Inspection Agency: this term designates the Inspection Bodies which work under the supervision of POGC. In some cases, Contractor nominated by POGC may also act as the Inspection Agency.

2. REFERENCE DOCUMENTS :

Replace : SP TCS 611 by : RP 6340S 999 6300 001.

3.1 Delete title and replace with "Materials subject to a post-weld solution annealing heat treatment (at 1150-1200°C) 3

3.2 Materials to be left in the as-welded condition:

Delete reference to "SAW process". 3

TOTAL EXPLORATION PRODUCTION

GENERAL SPECIFICATION

PIPING - VESSELS - WELDING

SP - TCS - 615

WELDING OF UNS 31254 ALLOY

REV. N°	DATE	PURPOSE OF THE REVISION
Rev. 2	February 1996	New issue
Rev. 1	March 1987	New issue
Rev. 0	July 1985	First issue

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1. SCOPE

This specification covers specific requirements for welding of materials (piping and equipment) made of austenitic stainless steel materials of the UNS S 31254 grade.

2. REFERENCE DOCUMENTS

The present document comes in addition to the general specification SP-TCS-611 : "welding of pressure containing piping and equipment" of which the applicable issue is the latest revision at the date of call for tender.

Any divergence between both documents, or between this specification and any other contractual document should be reported to TOTAL for decision. Should any discrepancy arise in using the present specification with any other contractual document, it is assumed that the more stringent requirement shall apply, unless otherwise agreed or decided by TOTAL.

3. WELDING PROCESSES

The following welding processes may be used :

3.1. Materials subject to a post-weld annealing heat treatment (at 1150 - 1200°C)

SAW
SMAW
GMAW
GTAW (with or without filler material)
PAW (with or without filler material)

3.2. Materials to be left in the as-welded condition :

SAW
SMAW
GMAW
GTAW)(acceptable only if used with filler material)
PAW)(acceptable only if used with filler material)

4. FILLER MATERIAL

The following commercial alloys are acceptable :

AVESTA P12
INCONEL 112
INCONEL 625

Any other commercial alloy of a similar composition may be used provided it meets the following requirements :

a) Chemical analysis (% weight)

$20 \leq Cr \leq 23$
$8 \leq Mo \leq 10$
Nb (as a stabilization element) : to suit the carbon content
Mn : ≤ 1.0
Fe : ≤ 7.0
Si : ≤ 0.75
Ni : Balance (≥ 50.0)

b) Mechanical properties

UTS ≥ 650 MPa
YS ≥ 300 MPa
Elongation $\geq 35\%$

5. PREHEATING AND INTERPASS TEMPERATURE

Preheating shall be avoided ; interpass temperatures should be kept to a minimum and shall not exceed 150°C. Metal shall be allowed to cool to about 100°C before the next pass is welded.

6. HEAT TREATMENT

No heat treatment shall be performed after welding, with the exception of a solution annealing at 1150 - 1200°C and only if required by contractual documents such as codes or standards (applicable to small pieces such as fittings, tubes, etc., only and not to fabricated spools, or vessels).

7. WELDING TECHNIQUE

- Single pass welding is not allowed.
- Single sided welding root passes shall be performed with the GTAW process and back purging exclusively.
- Heat input shall be kept to a minimum, and if possible not over 10 kJ/cm (15 kJ/cm maximum).
- Weaving shall be avoided.